

CSS Frameworks

Style React Apps Efficiently

■ LEARNING OUTCOME

By the end of this module you will style React apps with Tailwind CSS, use CSS Modules, apply styled-components, and use shadcn/ui for production-ready accessible components.

■ Skills You Will Gain

- Use Tailwind CSS utility classes
- Implement responsive design with breakpoints
- Use CSS Modules for scoped styles
- Apply styled-components for dynamic styles
- Use shadcn/ui components
- Choose the right styling approach

■ Choosing the right CSS strategy saves weeks. Tailwind CSS is used by 40%+ of modern React projects. shadcn/ui gives you production-quality accessible components in minutes.

SECTION 1

Tailwind CSS

```
# Install npm install -D tailwindcss postcss autoprefixer npx tailwindcss init -p #
tailwind.config.js: content: ['./src/**/*.js,jsx,ts,tsx'] # src/index.css: @tailwind base;
@tailwind components; @tailwind utilities; // Tailwind in React function ProductCard({ product }) {
return ( <div className='flex flex-col gap-4 rounded-2xl bg-white shadow-md p-6 hover:shadow-xl
transition-shadow duration-300 border border-gray-100'> <img src={product.image} alt={product.name}
className='w-full h-48 object-cover rounded-xl' /> <h3 className='text-lg font-semibold
text-gray-900 line-clamp-2'> {product.name} </h3> <div className='flex items-center
justify-between'> <span className='text-2xl font-bold text-violet-600'> Rs.
{product.price.toLocaleString()} </span> <span className='px-2 py-1 rounded-full text-xs
font-medium ${ product.inStock ? 'bg-green-100 text-green-700' : 'bg-red-100 text-red-700' }`>
{product.inStock ? 'In Stock' : 'Out of Stock'} </span> </div> <button className='w-full py-2.5
bg-violet-600 text-white font-semibold rounded-lg hover:bg-violet-700 active:scale-95
transition-all duration-200 disabled:opacity-50' disabled={!product.inStock}> Add to Cart </button>
</div> ); } // Responsive grid <div className='grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3
xl:grid-cols-4 gap-6 p-4'>
```

SECTION 2

CSS Modules, styled-components, and shadcn/ui

```
// CSS MODULES -- scoped class names // Button.module.css // .button { padding:10px 20px;
border:none; border-radius:6px; cursor:pointer; } // .primary { background:#7C3AED; color:white; }
// .primary:hover { background:#6D28D9; } import styles from './Button.module.css'; function
Button({ variant='primary', disabled, children }) { return ( <button className={[[styles.button,
styles[variant], disabled && styles.disabled].filter(Boolean).join(' ')}
disabled={disabled}>{children}</button> ); } // CSS Modules transforms .primary ->
Button_primary__abc12 -- globally unique // STYLED-COMPONENTS import styled from
'styled-components'; const StyledBtn = styled.button` padding: 10px 20px; background: ${p =>
p.variant==='danger' ? '#ef4444' : '#7C3AED'}; color: white; border:none; border-radius:6px;
cursor:pointer; &:hover { filter: brightness(1.1); } `; <StyledBtn
variant='danger'>Delete</StyledBtn> // SHADCN/UI npx shadcn-ui@latest init npx shadcn-ui@latest add
button card input dialog toast import { Button } from '@components/ui/button'; import { Card,
CardContent, CardHeader, CardTitle } from '@components/ui/card'; function ProductForm({ onSave })
{ const [name, setName] = useState(''); return ( <Card className='w-full max-w-md'>
<CardHeader><CardTitle>Add Product</CardTitle></CardHeader> <CardContent> <Input value={name}
onChange={e=>setName(e.target.value)} placeholder='Product name' /> <Button
onClick={()=>onSave({name})} className='mt-4 w-full'>Save</Button> </CardContent> </Card> ); }
```

■ PRACTICE Q & A

Q1. What is the difference between Tailwind and Bootstrap?

A: Bootstrap: pre-built components with fixed design. Tailwind: utility classes only -- no components, total flexibility. Tailwind is more flexible; Bootstrap faster for standard designs.

Q2. What are CSS Modules?

A: Automatically scope class names to the component file -- .button becomes Button_button__abc12. Eliminates class name collisions without naming conventions.

Q3. What is shadcn/ui?

A: Copies component source into your project instead of installing a package. You own the code -- fully customisable. Built on Radix UI (accessible) + Tailwind.

Q4. When would you choose styled-components over Tailwind?

A: For: highly dynamic styles based on many JS props, theme-driven design systems, teams comfortable with CSS syntax.
Tailwind for: rapid development, consistency, small bundle.

Q5. What is the cn() utility?

A: Merges Tailwind classes intelligently, handling conflicts. From clsx + tailwind-merge packages. Prevents duplicate/conflicting utility classes.

■ CSS Frameworks Cheat Sheet	
<code>flex items-center gap-4</code>	Flex row with centred items
<code>grid grid-cols-3 gap-6</code>	3-column CSS grid
<code>sm:grid-cols-1 lg:grid-cols-3</code>	Responsive breakpoints
<code>p-4 / px-4 py-2</code>	Padding: all / horizontal+vertical
<code>text-xl font-bold text-gray-900</code>	Typography utilities
<code>bg-violet-600 hover:bg-violet-700</code>	Background + hover
<code>rounded-xl shadow-md</code>	Border radius + shadow
<code>transition-all duration-300</code>	CSS transition
<code>import styles from './X.module.css'</code>	CSS Modules
<code>styles.className</code>	Apply scoped class
<code>npx shadcn-ui@latest add button</code>	Add shadcn component
<code>disabled:opacity-50</code>	Tailwind state variant